

Graphics Cards

A graphics card is what renders the image that you see onscreen. It's like a CPU, it has to process and execute a lot of commands albeit these are fewer in number than what a CPU has to handle. The job is repetitive in nature, so graphics cards have evolved to handle a lot of calculation of the same type. A CPU can handle the same workload but it'll be so overbearing that the entire computer will slow down to a crawl depending upon the workload. So now we have a system where the entire graphics workload is taken away from the CPU altogether and is managed by the GPU alone. Almost all CPUs released nowadays have an integrated graphics processor. This takes away the entire responsibility from the CPU, this way spikes in graphical processing will not hinder the entire computer. There are two major players in the desktop graphics market - NVIDIA and AMD.

Construction

The primary components of a graphics card include

1. The GPU chip
2. Video memory
3. RAMDAC
4. Power supply
5. Cooling

The focus on the power supply and the cooling are a rather new thing since earlier cards didn't get quite as hot and the act of overclocking GPUs wasn't that widespread. The GPU is where all the magic happens, programs send their calculations and this chip carries them out. The image created is sent to the RAM from where it will go directly to your monitor (if you use a digital interface like DVI or HDMI) or through the RAMDAC (converts digital image into an analog format which can be sent via the VGA port) to your monitor.

Like we talked about the power phase design in the motherboards section, graphics cards too have a power phase design. This is partly because the voltage levels that these work at are very low and the immense current consumption produces a lot of heat. Graphics cards that were meant to be overclocked have been seen with around 32 power phases, that is quite a lot.

Given the heat produced during heavy load almost all cards have an active heatsink. the more exotic versions come with 2-3 fans and copper heat pipes. There are a few common configurations used in designing the cooling system of a graphics card. One is to pull cool air from outside